Received: 6 March 2023 | Revised: 12 June 2023 | Accepted: 22 July 2023

DOI: 10.1111/tgis.13094

RESEARCH ARTICLE

Transactions in GIS WILEY

MapSafe: A complete tool for achieving geospatial data sovereignty

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Funding information
Ministry of Business, Innovation and Employment (Ngā Rākau Taketake – Myrtle Rust and Kauri Dieback Research, C09X1817)

Abstract

Sensitive geographic data are invaluable assets for the people to whom they belong and their disclosure should be decided by the sovereign data owner (SDO). Due to several high-profile data breaches and business models that commercialize user data, the need for new approaches to geoprivacy and data sovereignty has grown. We propose MapSafe, a client web application that first obfuscates datasets using donut masking or hexagonal binning, separately, and thereafter implements a multi-level encryption scheme that permits SDOs to share the final encrypted volume containing the geospatial information when they choose and at a level of detail which they are comfortable. The authenticity verification of the volume is facilitated by storing the hash value corresponding to the encrypted volume immutably on the Blockchain as a public record. Our approach places geoprivacy under data guardians'' control, and its integration capabilities promote its adoption in existing and future geospatial web systems.

1 | INTRODUCTION

Hosting spatial data in centralized databases and services has ballooned over the last decade, and an increasing number of people and organizations have adopted offshore and cloud-based models of data storage. However, although cloud solutions have enabled mass storage of data at affordable prices, they have concentrated control over these services to a handful of powerful companies (so-called *GAFAM*: Google, Amazon, Facebook, Apple, and

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Microsoft) (Calzada, 2021). These companies who own the technology can track and exert power over the data, resulting in rightful custodians of data, referred to as *sovereign data owners* (SDOs) in this article, unable to fully control their data (Elwood & Leszczynski, 2011). This has not only raised concerns around offshore data security and privacy but also the legal and privacy frameworks that the data are subject to, including the issue of data sovereignty (Kukutai & Taylor, 2016).

Data sovereignty has been at the core of various research activities over the last few years, especially in Europe. The topic gained additional traction through various regulations and initiatives such as the *EU General Data Protection Regulation* (GDPR), the *European Cybersecurity Certification Scheme for Cloud Services* (EUCS) and *Gaia-X European Association for Data and Cloud* (Gaia-X). With the increased utilization of cloud-based services, ensuring data sovereignty is becoming more important. However, little attention is paid to the law in those countries where data is stored. In contrast to the movement for greater data sovereignty in some nations, the US Patriot Act enshrines the government's right to access any data at any time and forbids the disclosure of that access (Barrett, 2018). The reforms of GDPR are changing this relationship, however (Barrett, 2018), and although GDPR is local to the EU, it has had cascading effects in the US (Chen et al., 2022; Klar, 2020).

Maps have historically been used for creating certain power and authority imbalances (Harley, 1989). Equivalent data sovereignty issues exist today due to data harvesting and increased surveillance due to big data, geoweb, and the cloud. Geospatial data are also increasingly stored in third-party services (such as one of GAFAM) due to the cost and reliability of cloud solutions. This approach nonetheless assumes a third-party trust relationship that enforces a reliance on others. As data increasingly flows across borders, data sovereignty has become a complex issue even for governments (Johnson & Scassa, 2023). Geospatial data sovereignty requires data to remain within the jurisdiction of origination and to prove territorial ownership. Settling these disputes via counter-mapping using historical geospatial datasets as evidence has been part of informal settlements and formal litigation (Gilbert & Begbie-Clench, 2018; Wainwright & Bryan, 2009). Since data could have been modified proving the authenticity of data is critical. During litigation, map authors can be required to disclose how geospatial data had been stored and secured from alterations (Fadaie, 2001). Thus, sovereignty and veracity over geospatial data should be provable in a court of law. They are even harder to prove in cloud environments since data and its control have been transferred to a third party.

Blockchain solutions are progressively being used for notarising and verifying the authenticity of geospatial data. Researchers (Sladić et al., 2021; Zhang et al., 2020; Zhao et al., 2022) have proposed approaches where geospatial data is stored on a decentralized file system, while a corresponding hash of the data is stored on the blockchain. The immutability of the blockchain has massive potential to resolve the limitations of the existing information systems concerning trust friction, transparency, and automation. These potential permit claims over the ownership and originality of the geospatial data to be resolved during informal and formal litigation, and have led to heavy investment in blockchain-based land administration schemes in several countries (Shuaib et al., 2020). However, geospatial data privacy entails additional challenges, and most solutions are limited in providing SDOs with a complete approach to protect these data without relying on a third party.

Geospatial data sovereignty is a multi-faceted challenge, requiring solutions to address multiple security concerns, especially regarding confidential storing and sharing of geospatial data. Often users with varying degrees of trust need access to different representations of the data—some audiences require access to high-precision data, while others do not. The same point is true for attribute data—some users require access to all attribute data values, and the richness of data contained therein is not suitable for all audiences. Geospatial data obfuscation offers a potential solution to make partial representations of the data available to some untrusted users (Ajayakumar et al., 2019; Cassa et al., 2008; Curtis et al., 2006; Swanlund et al., 2020; Zandbergen, 2014; Zimmerman & Pavlik, 2008). In environments where data are accessed by a variety of users, some of whom are less trusted than others, a range of cryptographic tools exist that ameliorate the concern of binary on–off access models, and allow us to protect the confidentiality, integrity, and authenticity of a given dataset. These methods have been shown to introduce sufficient entropy to ensure that data are difficult to reverse engineer and produce a hash value that

verifies if an encrypted file has been manipulated or tampered with after creation. However, this approach also suffers from limitations as both the encrypted file and its hash can be replaced with false ones (Meyer & Wetzel, 2004). A solution that immutably enshrines the hash is a key part of this challenge. In addition, most contemporary geoprivacy solutions do not empower users with systematic and detailed control over access to geospatial data when they are being shared, as these tend to rely on simple binary classifications (e.g., on/off) (Liu et al., 2016; Martin & Nissenbaum, 2020). In cloud-based geospatial applications, accommodating user-control and ownership over geospatial data requires consensus-driven data storage that can be verified and trusted. In doing so, there may be the potential to adopt a trustless approach to geoprivacy protection during data sharing.

Essentially, these challenges require a geoprivacy solution that provides SDOs with a sharing facility whereby they can store the encrypted data either on their devices or on the cloud, and still be able to share them appropriately with SDPs with varying levels of trust, without compromising security. The approach should also embed a means to make a claim to or reference a dataset without exposing the contained coordinates, and one which intended SDPs can also use to verify data authenticity. Implementing such a security tool has challenges, the largest of which are the necessary features that ensure the process is accessible for non-experts (Ajayakumar et al., 2019), and to attract users that already have established usage patterns (Cranor & Buchler, 2014; Möller et al., 2011; Yee, 2004). Although such a tool has the potential to change data sovereignty, implementations are yet to emerge.

This article presents *MapSafe*, a complete application designed to address these challenges. The browser-based tool lets SDOs first obfuscate sensitive geospatial dataset using geo-masking or binning separately. The original and obfuscated datasets are further protected via a multi-level encryption scheme that accommodates varied trust levels of SDPs during sharing. Finally, for authenticity verification, a hash value of the resulting encrypted file is notarised on the tamper-free blockchain. The key idea is to enable SDOs to keep sensitive geospatial data on their computer and allow them to perform all security safeguards locally without the need for a third party.

The remainder of this article is structured as follows. The next section describes the concerns and attempts in realizing geospatial data sovereignty in the literature. Section 3 describes the structure and components of the MapSafe tool, including the technical details and performance evaluation. Section 4 analyses the security provided by MapSafe, while a discussion on its implications, limitations, and future directions for geospatial data sovereignty research are presented in Section 5. The article provides our conclusions in Section 6.

2 | CURRENT GEOSPATIAL DATA SOVEREIGNTY CONCERNS

Data sovereignty has been described as the "management of information in a way that aligns with the laws, practices and customs of a nation-state in which it is located" (Lovett et al., 2019, p. 27). Data sovereignty has become a significant challenge due to the advent of big data and the rise of transnational tech giants (Lukings & Habibi, 2022). The volume, variety and velocity of big data (Kitchin, 2013) create a necessity for flexible and layered data storage and sharing intermediaries that result in data being stored in locations independent of national boundaries (Amoore, 2018; Hummel et al., 2021). This has prompted some countries to take steps to safeguard and protect sensitive national data (Nugraha et al., 2015). For geographic data, early examples include government protections surrounding the public's faces on Google Street View (Alvarez León, 2016), though modern protections have focused on transnational data transport, resulting in the Global Data Protection Regulation (GDPR) (EU Regulation, 2016). These regulations recognized that individuals and groups outside of the public sphere have a right to data privacy, including control of how information is held, transferred, and which actors have access (Hummel et al., 2021; Kukutai & Taylor, 2016). This goal, however, is not so easy to realize, and carefully crafted solutions are necessary.

On many occasions, geospatial datasets include critical or culturally sensitive locations, such as protected environmental sites, which should not be disclosed to the private corporations trying to profit from these data (Gómez et al., 2022; Johnson & Scassa, 2023; Quach et al., 2022; W3C, 2022). Nevertheless, most companies do not have

anywhere near the resources to protect their infrastructure such as hyperscalers AWS, GCP, or Azure, and are therefore adopting their cloud-based solutions. This inherently brings an increased risk of security breaches compared to traditional on-premise data storage. In the last 4 years, security breaches on the cloud have impacted major companies, including Facebook, Alibaba, and LinkedIn, exposing millions of records—the majority including location information (Arcserve, 2022). While these breaches were from unknown adversaries, individuals within corporations can gain unauthorized access to data. This section describes the various facets of geospatial data privacy and confidentiality.

2.1 | Geospatial data privacy and confidentiality preservation

Geomasking is the most widely used and studied geospatial data privacy-preserving technique (Ajayakumar et al., 2019; Polzin & Kounadi, 2021; Swanlund et al., 2020). Geomasking approaches, such as donut masking, displace point data within a certain distance threshold, with the aim to create masked datasets that continue to enable meaningful analysis (Armstrong et al., 1999; Duckham & Kulik, 2006; Zandbergen, 2014). Reid and Sieber (2020) highlight that while significant developments have been made in geographic masking techniques (Cottrill, 2011; Kwan, 2004), Geoweb technologies have not incorporated these techniques as part of their solutions to reduce the risks in data sharing. Masking, however, is insufficient for complete security because locations contained in masked data can sometimes be reverse-engineered (Keßler & McKenzie, 2018; Seidl et al., 2018; Zhang & McKenzie, 2022). Indeed, Zimmerman and Pavlik (2008) and Cassa et al. (2008) have demonstrated that original locations can be estimated when multiple versions of the same dataset are released using the same masking methodology.

Binning is an alternative technique used to anonymise geospatial datasets where points (locations) within a region are aggregated. This is useful for scenarios when only the density of points needs to be conveyed, that is, answering the *how much?*, rather than *where?* In *Hexabinning* for instance, a grid of hexagons is layered over a two-dimensional grid, and the data in every hexagon is counted and labeled to classify the contained data. Hojati et al. (2021) employ Uber's H3 hexagonal hierarchical spatial index binning (http://eng.uber.com/h3/) as part of their geoprivacy framework for the distributed web. This provides an explicit level of uncertainty and geo-masking embedded into each of the cells in their grid, and enabled geographic information sharing with different levels of masking or uncertainty without revealing coordinates. Rao et al. (2021) recently proposed a privacy-preserving location recommendation framework using H3 to aggregate and encode the latitude and longitude of people's locations. Binning, however, also has limitations. Studies highlight that aggregation performed to fine units only provides limited anonymity (De Montjoye et al., 2013), and useful information can be computed (Xu et al., 2017). Increasing to a coarser level of aggregation certainly resolves this, but results in a loss in spatial resolution, resulting in limited utility (Gao et al., 2019).

Considering the privacy limitations with geomasking and the difficult trade-off between privacy and utility with binning, further means of protecting geospatial datasets are necessary. Furthermore, beyond coordinate data, spatial metadata, and attributes hold significant information, which is not always appropriate to include with a masked dataset. What is happening at a location can be more sensitive than where and research to date has not focused on this issue. In these cases, additional truncation or encryption of these fields is vital to provide an extra layer of data protection.

Encryption is a common approach employed to protect data against unwanted disclosure and is a critical aspect of data sovereignty and spatial data sharing. Geospatial file data encryption approaches include those for Bitlocker drive (Harris et al., 2021), as a plugin in GIS software (Ghaleb et al., 2017), or custom algorithms that selectively encrypt vector-based spatial files (Wang et al., 2021), while conventional database-based spatial data encryption has been proposed by Li (2010). Similarly, Cloud-based geospatial data protection approaches have included encryption as part of their offerings. Esri's ArcGIS Online (Esri, 2022), Microsoft Azure Orbital's offering for geospatial data (Microsoft, 2022), and Google Bigquery (Google, 2022) all offer solutions for geospatial data encryption at rest. However, SDOs do not always trust CSPs since the secret keys used to provide security are now under their control.

To this end, researchers have proposed encryption schemes for data before their storage on the cloud, such as Abroshan (2021). Nevertheless, current geospatial data security approaches are still limited in provisioning with whom, and how much, the encrypted data is to be shared (Zhao et al., 2022). This intrinsically calls for arming SDOs with an encryption scheme that enables them to safeguard and selectively share their data by themselves.

Imparting control to the SDOs necessitates security functions implemented in a way that no sensitive data leaves users' computers. The most closely related work in this regard is by McKenzie et al. (2022) whose *PrivyTo* tool enables users to obfuscate, encrypt, and publish locations and share them using their devices. The tool furnishes a uniquely open method for sharing an individual's location. A complete data sovereignty solution, however, requires provisioning geoprivacy for complete geospatial datasets, and needs to accommodate the secure sharing of reduced representations of original geospatial data in response to varying user trust levels, as will be explained in Section 2.3. To the best of our knowledge, there are currently no works that focus on resolving these challenges.

2.2 | Geospatial data integrity and verification

Though encryption provides a means to secure geospatial data on the cloud, there still remain vulnerabilities. Using encrypted datasets also has the potential to enable person-in-the-middle style attacks (Conti et al., 2016; Meyer & Wetzel, 2004) where the encrypted data is replaced or altered without the recipient's knowledge. To combat this, alongside the encrypted file its corresponding unique fingerprint, a *hash value*, can be published that an SDO can use to prove its originality, and of the geospatial dataset within, without needing to reveal coordinates. However, an adversary can replace both the encrypted file and its corresponding hash. If these hash values are to be used for verification that encrypted data have not been tampered with, it is imperative that it is unassailable. Anything less is insufficient to ensure that encrypted volumes are who they purport to be. Ideally, hashed values of encrypted volumes would be stored in a public fashion, so that they could be easily verified.

To combat this challenge for geospatial datasets and location identifiers, researchers and industrialists are increasingly leveraging decentralized notarisation approaches. Ethereum and other Blockchain technologies (referred to as the blockchain in this article) offer a mechanism capable of forming zero trust sharing, eschewing the need for centralized data brokering authorities, such as GISCloud (giscloud.com), Esri ArcGIS Online (arcgis.com), or Amazon Web Services (aws.amazon.com). Geospatial blockchains are offering traceability and unforgeability in several aspects: farming-related data (Miloudi et al., 2019), land administration (Sladić et al., 2021), and highly sensitive site selection (Farnaghi & Mansourian, 2020). An example that encompasses these works is by Tai et al. (2021) who present a blockchain-based spatial data trading platform to overcome dishonesty in the trading process. Sellers first encrypt the data, upload it to the *Interplanetary File System* (IPFS), and then register a corresponding hash value data on the blockchain. Buyers use this hash for verification while bidding on these data. Furthermore, several countries have initiated projects migrating to blockchain-based land registries in their efforts to save time and reduce the opportunity for bribery and fraud (Benbunan-Fich & Castellanos, 2018; Shuaib et al., 2020).

2.3 | Accommodating different trust levels

Data sovereignty solutions are incomplete without giving SDOs the ability to exercise their rights and control the usage and sharing of their data (Hojati et al., 2021; Zhang et al., 2020). As such, accommodating sharing of potentially sensitive data with different sets of users based on varying degrees of trust is a key prerequisite and encourages collaboration between otherwise isolated members. Geospatial researchers have recently emphasized that “fine-grained control over the resolution of spatio-temporal data being shared and with whom they are sharing is not provided by most geoprivacy approaches, which tend to rely on simple binary classifications” (Hojati et al., 2021, p. 2). The authors propose a geoprivacy solution that gives social media users control over how their

locations are disclosed (higher or lower precision) to their connections based on social trust, namely, the length of their friendship on Twitter. Their locations are obfuscated using hexagonal binning, and users are subsequently allowed to encrypt and host it on *IPFS* or their machines and for sharing it upon request. While they also focus on data sovereignty, their work, like McKenzie et al. (2022), falls in the area of secure individual location sharing, not for protecting geospatial datasets. Additionally, not all sovereign data parties (SDPs) may be connected via social networks. For sharing data with users with varied trust levels, it is essential that a data sovereignty solution accommodate different encryption levels per the corresponding data granularity.

In summary, while scholars have begun to research the utility of data sovereignty technology for geospatial data, there are few, if any, fully realized tools that address these challenges, especially one that can be applied using a limited geospatial skillset. Most geospatial data privacy and integrity implementations have traditionally been designed for centralized data sets that required sharing data with a central server in some way to perform the security safeguards. Standalone works focusing either on geomasking, encryption, or blockchain notarisation, fall short of providing complete data sovereignty. The next sections describe our tool that addresses these challenges.

3 | MAPSAFE.XYZ: ARCHITECTURE AND IMPLEMENTATION

We propose MapSafe, a complete data sovereignty application for safeguarding and verifying sensitive geospatial data. The client-side web application (https://www.mapsafe.xyz) allows SDOs to perform geomasking, encryption, and blockchain notarisation resulting in a complete approach to data sovereignty. SDPs can use the blockchain notarised identifier to verify the authenticity of the data before decrypting and viewing it. We demonstrate MapSafe's features through user *Jane*, who represents an SDO working for an Oil and Gas company needing to protect the privacy of newfound potential drill sites but also share and notarise this data. These sites around North Coromandel in Aotearoa New Zealand need to be shared with her supervisor *Bill*, who lives across the world and needs to view the actual drill sites, while other SDPs with lesser privileges need to view only an obfuscated representation.

For providing complete geoprivacy, MapSafe's architecture (Figure 1) has two main components: a data privacy front-end and a blockchain back-end. Jane can perform the safeguarding functions within the web browser without any additional software, or even an Internet connection, once the webpage has been loaded. No sensitive data leaves Janes's computer unprotected, apart from the filename and the hash value of the encrypted geospatial

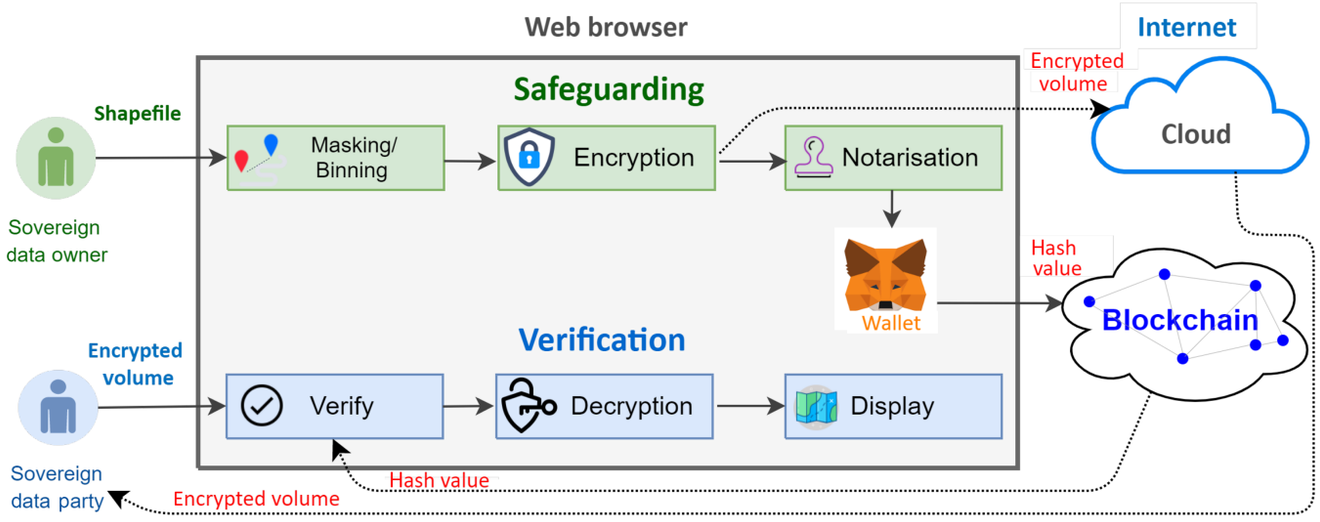


FIGURE 1 Simplified MapSafe's safeguarding and verification processes. An SDP loads a shapefile and performs masking or binning, encryption, and notarisation of the hash value on the blockchain. Recipients of the encrypted volume verify it using the hash before decrypting and displaying the dataset contained within.

volume for notarisation on the blockchain using a digital wallet, such as *Metamask* (https://metamask.io). Jane can share the encrypted volume with Bill either directly, via uploading to a Cloud server, or store it on her computer. Bill can first use the hash to verify the encrypted volume, decrypt it, and view the drilling sites. These safeguarding and verification aspects are described in more detail next.

3.1 | Geographic data obfuscation and accommodating different trust levels

As the first step, MapSafe enables SDOs to create obfuscated representations of the original dataset for sharing with lesser privileged SDPs. The MapSafe masking module employs the MaskMy.XYZ JavaScript library from Swanlund et al. (2020). The library performs donut masking, a technique that randomly perturbs each point in a dataset within a minimum and maximum distance threshold of the original point location, enabling SDO Jane to generate and share approximate locations of drill sites with users with lesser privileges. Figure 2 shows the masking performed on the North Coromandel dataset. This fictional dataset contains synthetically derived coordinates for demonstrating MapSafe's features. After loading a shapefile containing the dataset, Jane chooses the minimum and maximum masking distance parameters that the tool would use to generate a random location for transforming each drill site coordinate. For assistance in determining whether the chosen boundaries are adequate, we have included Spruill's measure (Armstrong et al., 1999; Swanlund et al., 2020). This measure indicates the percentage of masked points that are closer to any sensitive point other than their original point, making them

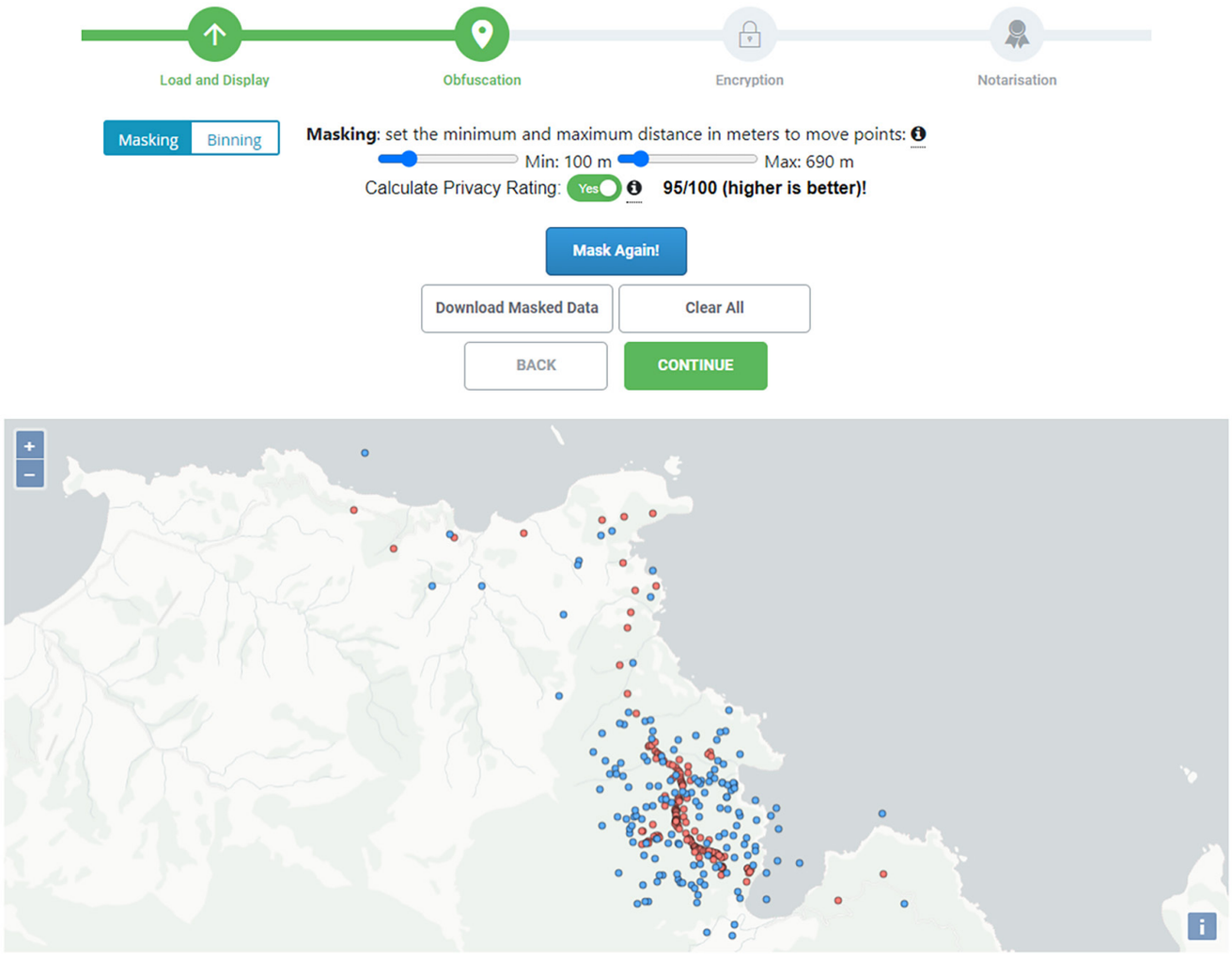


FIGURE 2 Donut masking the North Coromandel dataset. Drill sites are obscured by transforming their coordinates to a random location within the user-defined minimum and maximum bounds.

more vulnerable to re-identfication. In MapSafe, this function has been inverted so that a high number equates to a high level of privacy, which new users may find easier to grasp. Computing this measure in MapSafe is optional since it takes considerable time. MapSafe includes safeguarding and verification guides, including videos that demonstrate both processes from start to finish.

The masking times for shapefiles[1] are included in Table 2, with all tests performed using the Chrome browser on a computer with an Intel i5-8265U CPU 1.7 GHz processor with 8 GB memory running Windows 10 OS. Without Spruill's measure, on average 10,000 points are masked twice in a second. With this measure, on average 440 points are masked per second (based on a dataset of 10,000 points) and this average decreased with more points—100 points per second for a dataset with more than 60,000 points. This is likely due to the fewer computations and hardware paging required when the browser accommodated massive temporary data generated while computing this measure.

Hexagonal binning (*hexbinning*) is the other option for anonymising geospatial datasets in MapSafe. Geographic points are aggregated into hexagonal cells using Uber's h3-js library (https://github.com/uber/h3-js). Hexagons with more points are shown as darker blue cells, while those with fewer are shown as lighter blue cells. Jane can choose the spatial resolution (i.e., the Uber H3 spatial indexing level) along with the buffer radius for encoding. The buffer radius (in KM) allows specification of how far the coverage of the binning should span. Depending on the area depicted in the dataset, a suitable spatial resolution level will need to be chosen to balance privacy and utility. Large hexagons (lower resolution) would result in many distant locations shown in the same cell, while small hexagons (higher resolutions) may present results that are too sparse to cover the entire area (Rao et al., 2021). Therefore, a resolution size needs to be chosen that covers the entire area (i.e., dense representations) and contains a small cell size (i.e., preserving location details) to better balance the trade-off between spatial coverage and location details than other H3 indexing levels (Rao et al., 2021). Figure 3 shows the binning results for the North

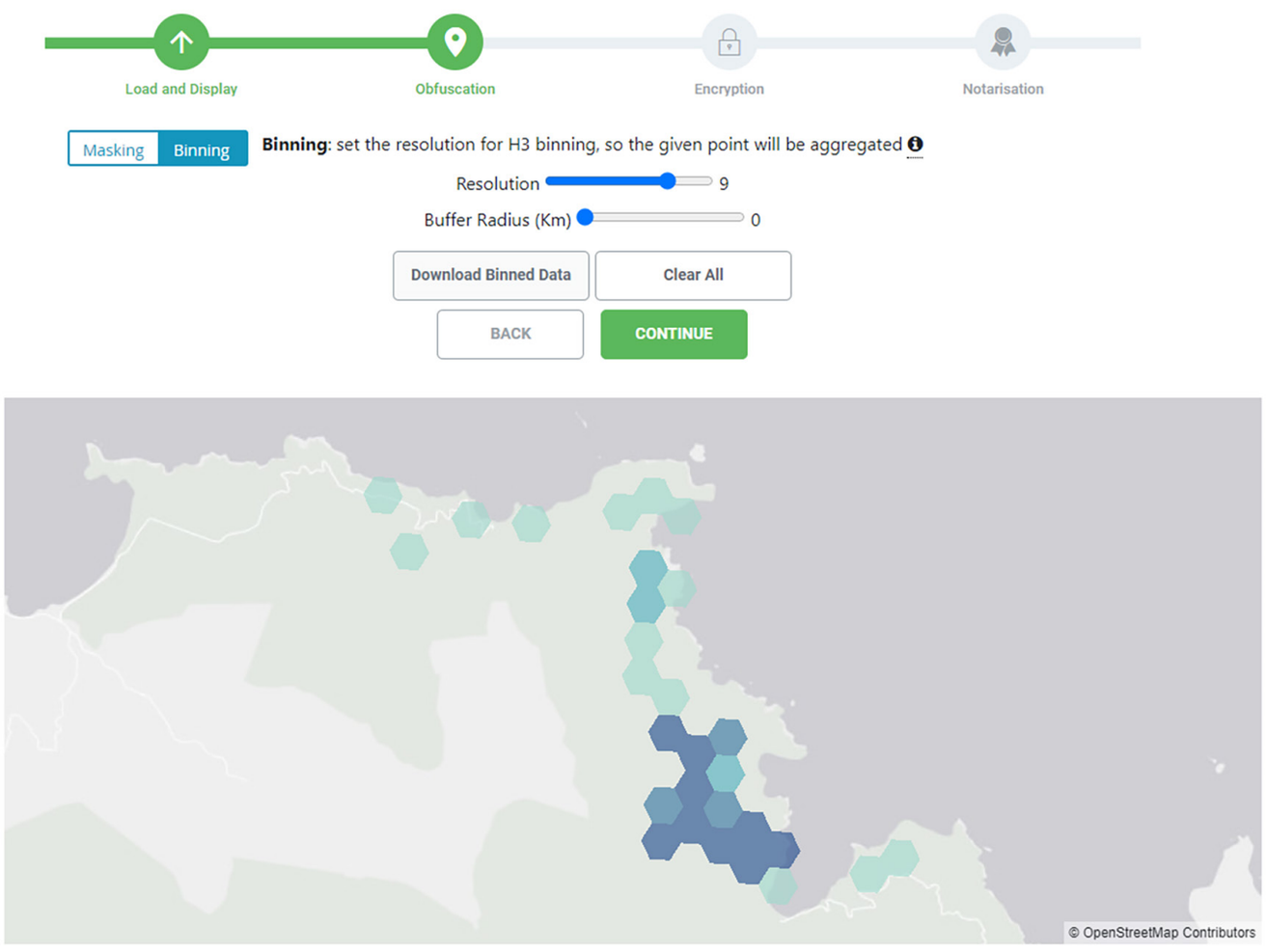


FIGURE 3 Hexagonal binning aggregates drill sites at different resolutions (9 chosen here) and buffer radius.

* FOR SACRED SITE!! BURIAL GROUNDS = WAKAN LATS/LONGS!!

Coromandel dataset for *resolution* = 9 (the level of a small city). The hexabinning process of generating hexabins and displaying them is instantaneous.

MapSafe permits Jane to choose different sharing options to accommodate varied trust levels. Trusted users, such as Bill, access the original locations. For semi-trusted users who are trusted to a certain extent, she can add a second level that will contain masked or binned locations. Here, the masking will use random numbers lying between the user-provided minimum and maximum bounds, while binning will use the user-provided H3 resolution and buffer radius. For untrusted users who are not trusted, she can add a third level that contains heavily masked or binned locations. Herein, masking retains the user-provided boundary difference but begins the boundary from the provided maximum. Binning uses one less user-provided H3 resolution and increases the provided buffer radius by 0.5 km.

The multiple obfuscation approach provides better privacy options and enables Jane to share datasets at appropriate levels. Figure 8 illustrates these original, masked and binned layers. After both masking and binning, Jane can download the resulting masked or binned dataset in shapefile format. During encryption, these user-defined parameters are added as part of the final encrypted volume to later give Jane an accuracy assessment of the original data. This is important in terms of being able to share the output, having a simultaneous interpretation between both parties (Jane and Bill), and being able to share aggregated or masked original surfaces without concern. While the masking feature only caters for points, line and polygon-based datasets can still be safeguarded in MapSafe by skipping masking and proceeding directly to encryption and further notarisation.

3.2 | Multi-level encryption

In MapSafe's encryption scheme, the original geospatial dataset alone or together with its obfuscated representations (*medium coarse* and *coarse*), are incorporated into a single encrypted volume that can then be shared with users who can decrypt them based on their privilege (detailed in Table 1). This encrypted volume contains multiple secret levels, each holding a different representation of the original dataset, in a nested encryption scheme.[2] Figure 4 shows the encryption feature, where SDOs can choose to encrypt only the original dataset, or together with its obfuscated representations. Jane can now share the encrypted volume with Bill, along with the full passphrase for him to decrypt into the inner level which holds the original dataset. The nested approach makes it easier for sharing and coordination. The parameters used for masking (or binning) at each level are stored one level inside.

Encryption is performed within the web browser using the SubtleCrypto module (Mozilla, 2022). The AES-GCM algorithm with 256-bit keys is used as it has better security compared to other modes of the algorithm as it first authenticates the encrypted content before decryption. The encryption and decryption times are shown in columns 6–11 of Table 2. These times include the duration required for zip file and shapefile conversion at each level of the volume. The encryption times at each level correspond to the three different scenarios: fine only,

TABLE 1 Based on their privileges, sovereign data parties are shared with different passphrases that enable access to different representations of the original dataset within the encrypted volume.

Sov. data party	Passphrase	Privilege to level	Dataset view	Anonymisation level
Trusted	15 terms	Inner	Fine (original)	None
Semi-trusted	10 terms	Middle	Medium coarse	Masked or binned
Untrusted	5 terms	Outer	Coarse	Heavily masked or binned

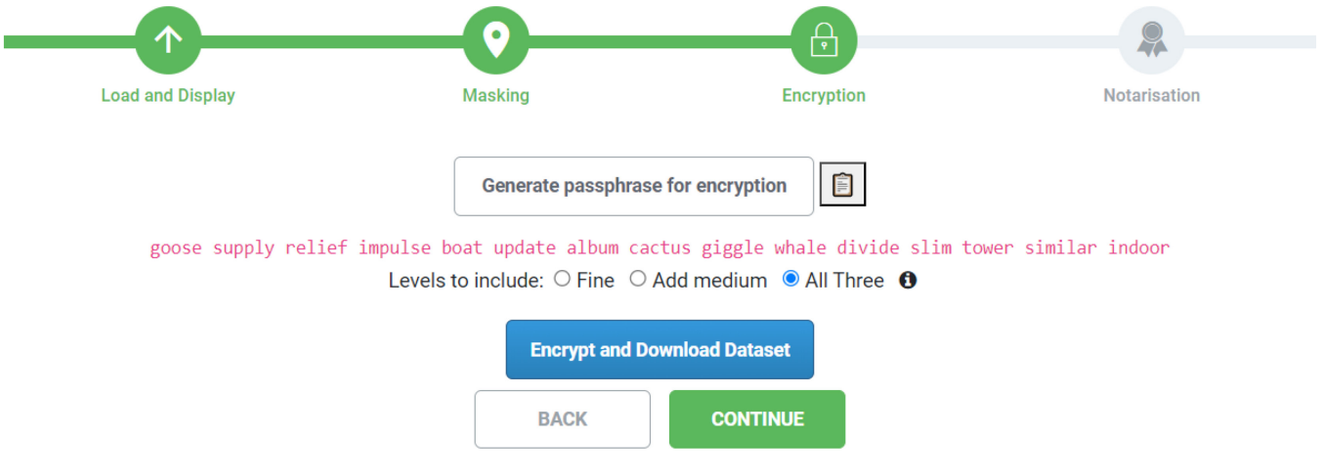


FIGURE 4 The encryption feature lets SDOs choose a passphrase and the different representations of the original dataset to include in the final encrypted volume.

fine & medium, and fine, medium & coarse, while decryption times at each level denote the time it takes to decrypt to the fine (first) level in each of these scenarios. For triple encryption, we see an average speed of about 6 s per mb (1024 kb), while decryption scores faster average speeds of about 0.8 s per mb. Similar to masking times, the encryption time is exponential rather than linear, as it increases with the file size and the number of levels. For twice the filesize (2 mb) the resulting time increases by 1.25 times since each subsequent encryption generates a larger file. Interestingly, encryption and decryption of one and two levels take almost the same time as for three levels. These encryption times are fast and practical, given that sensitive geospatial datasets may not be extremely large. The JavaScript libraries used in MapSafe are stored locally on the website, preventing any off-site calls from the browser. — PYTHON ?

Our approach provides SDOs and parties with flexible and sound schemes for sharing the symmetric key and protecting the encrypted volume. Jane can share the symmetric key with Bill using a secure channel, such as the Signal encrypted messaging app (Cohn-Gordon et al., 2020). Both Jane and Bill can safeguard this passphrase on their computers using a local password manager, such as KeePass Password Safe (KeyPass, 2022) or a built-in OS password manager without cloud backups, for example, the Apple iOS Keyring. They both can safeguard the encrypted volume on their computer, organization server, or in a cloud-based service. These options are valuable for safeguarding the encrypted original data, an essential artifact needed in case of authenticity verification during litigation.

TABLE 2 Shapefile masking, encryption, and decryption times (in *seconds*) in MapSafe using the Chrome browser. Masking is always performed twice. The encryption times correspond to the time taken to encrypt datasets at each level, while the decryption times denote the time taken to decrypt the fine level from each encrypted level.

			Masking		Encryption			Decryption		
Shapefile	Size (kb)	Points	w/o SM	w/ SM	1st	2nd	3rd	1st	2nd	3rd
Belgium	363	13,427	1.37	30.52	0.63	1.32	2.39	0.36	0.46	0.57
Sweden	587	20,873	2.19	77.00	0.64	1.62	3.37	0.37	0.50	0.65
Haiti	1010	28,581	2.83	131.05	0.70	2.38	6.20	0.38	0.54	0.82
Indonesia	1227	34,798	3.48	204.51	0.72	3.28	7.44	0.40	0.60	0.82
Philippines	2323	63,575	6.10	642.80	0.84	5.51	9.36	0.40	0.71	1.12

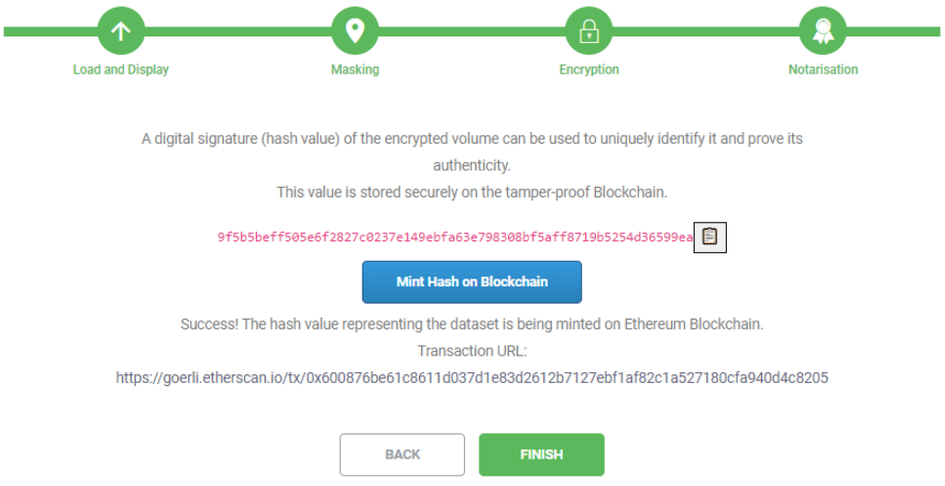
3.3 | Blockchain notarisation

In MapSafe, the filename and hash value of the encrypted volume containing the original and obfuscated geospatial datasets is notarised as a public record on the tamper-resistant Ethereum blockchain. Since all blockchain transactions are "public", the transactions and their data recorded in the ledger are visible and verified by everyone. This way, Jane can keep or share the encrypted volume, while its filename and hash value, stored as a non-modifiable public record on her blockchain address can always be used to check if the file has been altered—completing our geoprivacy approach.

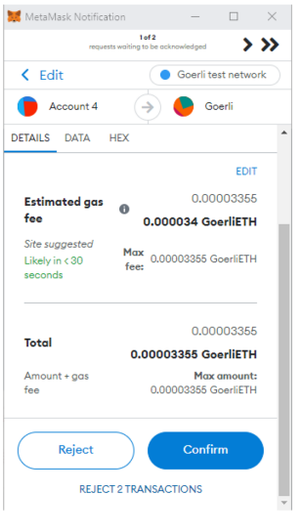
Non-Fungible Tokens (NFTs) introduced recently in blockchains are more suited to identifying ownership of assets or anything of interest or value. Storing the filename and hash value as an NFT on the blockchain furnishes a public token that uniquely identifies the encrypted volume holding the potential drill sites stored as a cryptographically unique, non-fungible digital asset within Jane's wallet, similar to how artwork is stored on a blockchain (Chevet, 2018). Bill can verify the originality of the received encrypted volume by matching its hash with the one attached to NFT. In case of disputes, Jane can do the same to prove the identity of the encrypted volume and therefore, the authenticity of the geospatial datasets contained within. We believe the hash value matching should suffice during litigation, without the need to decrypt the encrypted volume as this would disclose the coordinates. The only requirement is to ensure Jane, the SDO, retains the encrypted volume. This verification process is described in the next section.

Addresses and transactions are essential concepts in cryptocurrencies. Jane requires an Ethereum blockchain address for minting the encrypted volume's details, and a cryptocurrency wallet, such as *Metamask*, to connect to this Ethereum address for minting. The minting will append a new transaction to her address, storing these details as persistent evidence of the encrypted volume (see https://goerli.etherscan.io/tx/0x636b6e29f03f75bdf60c6a57625b79b7d4d4efd41b7d96ffc3e463384a134f22). This function requires sufficient Ether on Jane's account to pay for the transaction, which she can acquire via popular crypto-selling websites, such as https://www.coinbase.com.

Figure 5 shows the hash value of Jane's encrypted file being computed for notarisation on the blockchain as the last step. After pressing the minting button, the *Metamask* wallet prompts Jane for confirmation on the transaction, cost, and minting priority (*high*, *medium*, and *low*). Upon confirmation, the encrypted volume's filename and hash are submitted to Ethereum to be minted as an NFT and thereafter, forever existing as a



(a) The notarisation feature showing the file details being minted on the blockchain.



(b) The metamask wallet presented to the user.

FIGURE 5 As the final step, the encrypted file's name and hash value are notarised on the Ethereum blockchain.

public record. The transaction minting process is generally not instant and can take from a few minutes to a couple of hours to complete depending on the queue and the chosen priority. To assure users, a PDF invoice is generated that contains the transaction details, including the URL on Ethereum where the file details will be minted.

3.4 | Verification

MapSafe's verification process aims to prove the originality of the encrypted volume received by an SDP, based on the hash value stored on the blockchain. As shown in Figure 6, once Bill loads the received encrypted volume onto the browser, its hash value is computed and displayed for him to confirm against the original encrypted volume's hash minted along with the filename under Jane's blockchain address. Bill can then proceed to decrypt the volume, choosing *original* map granularity and providing the full 15 terms of the passphrase, as shown in Figure 7. After decryption, Bill would be presented with the original dataset. Other SDPs with lesser privileges, possessing lesser terms of the passphrase, would be presented with an obfuscated dataset from the level they have decrypted to. Figure 8 illustrates the different representations of masked datasets (left) and binned datasets (right).

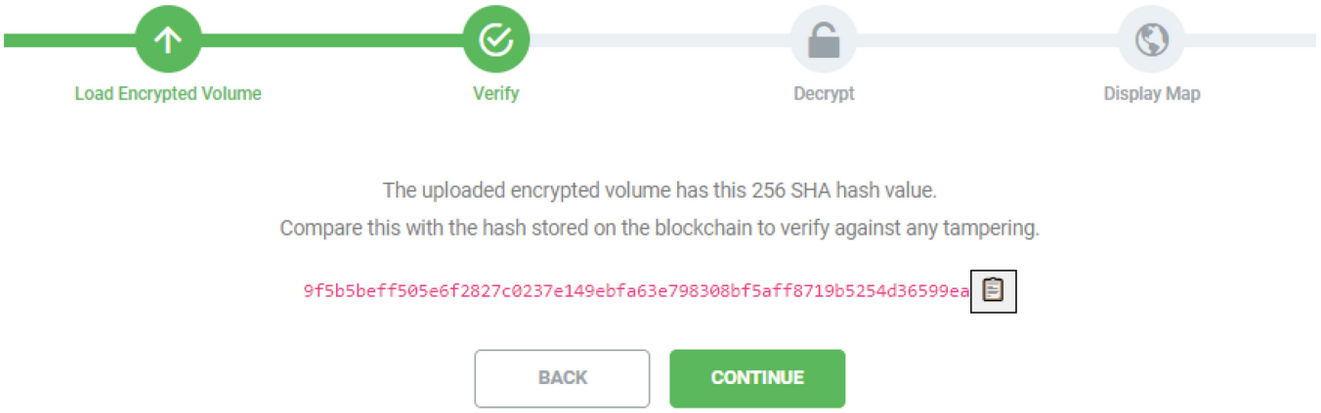


FIGURE 6 The verification feature allows SDPs to use the hash to verify the encrypted volume's originality.

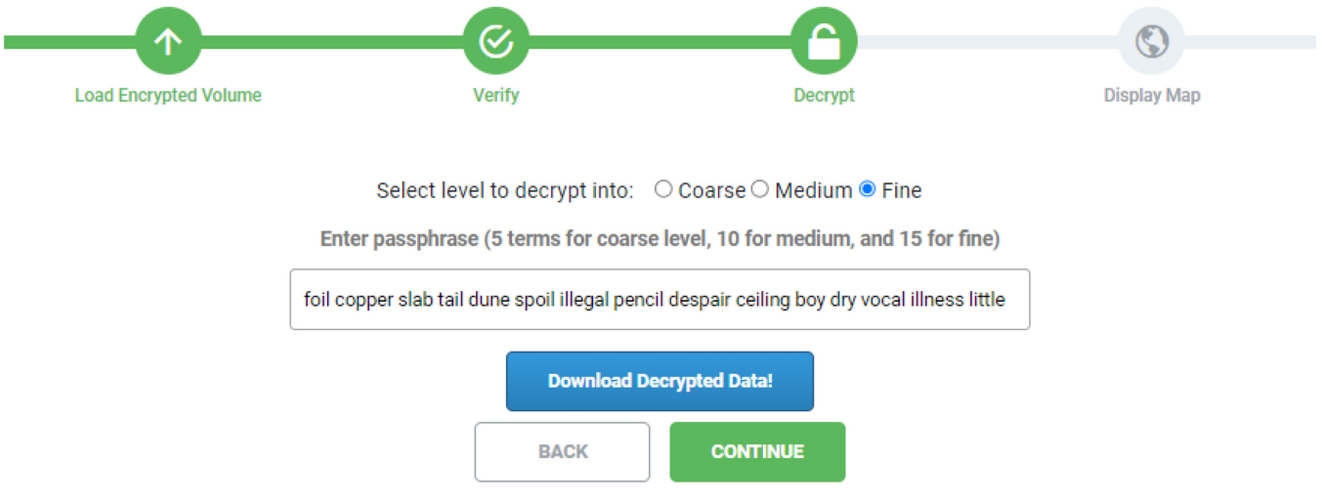
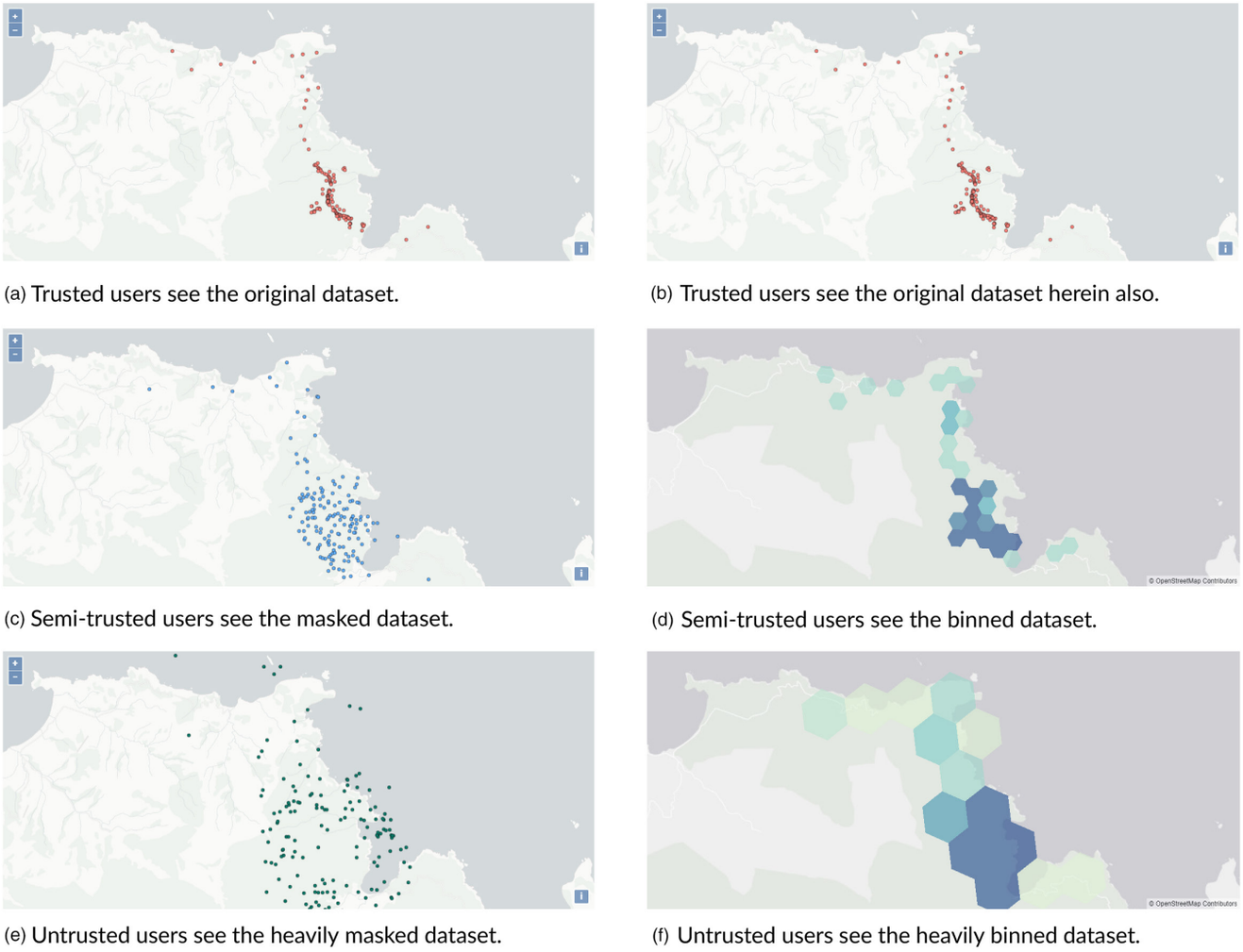


FIGURE 7 Different SDPs provide passphrases with varying lengths for decrypting to a specific encrypted level, and viewing the dataset contained therein as a map (see Figure 8).



(a) Trusted users see the original dataset.

(b) Trusted users see the original dataset herein also.

(c) Semi-trusted users see the masked dataset.

(d) Semi-trusted users see the binned dataset.

(e) Untrusted users see the heavily masked dataset.

(f) Untrusted users see the heavily binned dataset.

FIGURE 8 Masked (left) or binned (right) datasets are displayed after verification. Trusted users in both cases see the original (a) or (b), semi-trusted see medium coarse (c) or (d), while untrusted users see coarse dataset (e) or (f).

4 | SECURITY ANALYSIS

One of the biggest concerns with Internet-connected devices is trust, and this breaks down into two aspects â€ “the trust of the user in our system, and the trust that the blockchain itself is not manipulated. First, users must trust MapSafe regarding how their data are used. Without total transparency, users and organizations could be led to believe that they are interacting with a secure platform when, in fact, they are not. Our tool eliminates the need to send sensitive geospatial data to a central server for security safeguards by implementing all security functions locally within the web browser. This arrangement lets SDOs keep the encrypted data or share it via third-party services, such as cloud storage and ensures that at no point are the data managed or accessed by anyone other than the sovereign parties. Second, using a blockchain for notarising the encrypted volume's filename and hash value as a public record for verification furnishes a trustless system that is not controlled by a third party. Hence, this verification information is tamper-free, immutable, and publicly accessible to institutions and citizens. In this way, we provide a complete approach for enabling sovereignty over geospatial data in cloud-based environments.

When Jane decides to share the encrypted volume with Bill over the network or to store it on the cloud, two types of adversaries can try to break the security and obtain useful information. The *passive attacker* can eavesdrop on the web communication channel. However, upon interception, the encrypted data cannot be decrypted without the passphrase. Using a brute force approach for decryption would be prohibitively expensive

considering the security of the AES algorithm and the 15-term passphrase. The *active attacker* will try to impersonate the MapSafe website or trusted users by forging or replaying messages to fool users into believing they are valid and thereby leaking them passphrases (person-in-the-middle attack). Following our motivation, we hope users will be vigilant when storing their passphrases and verify the real identities of websites and individuals requesting the passphrases. Undoubtedly, data sovereignty tools are not a silver bullet. They can only be effective if they are used in conjunction with other privacy-enhancing approaches, such as securing the client's operating system, firewalls, and sound key management.

5 | DISCUSSION

Moving from on-premise to cloud-based data storage brings about an immense potential for geospatial data, but ameliorates geoprivacy concerns against data harvesting assemblages. Obtaining sovereignty over geospatial data, however, is a multifaceted challenge. Current geoprivacy approaches have only addressed one or more of these challenges: focusing on masking (Ajayakumar et al., 2019; Hojati et al., 2021; Swanlund et al., 2020), hexagonal binning (Hojati et al., 2021; Rao et al., 2021), encryption (Hojati et al., 2021), or notarisation on the blockchain (Farnaghi & Mansourian, 2020; Sladić et al., 2021; Zhang et al., 2020), and/or those that offer multi-level geomasking with encryption (Hojati et al., 2021). *MapSafe* leverages all these security functions in a workflow to offer a more holistic and complete geospatial data sovereignty approach. SDOs can obfuscate their geographic data and, in doing so, control the level of obfuscation for sharing them with SDPs. Geomasking alters coordinates preserving the spatial patterns while binning discloses the densities of locations only. The obfuscated and original datasets are then encrypted based on different trust levels applied in the browser and offline, only after which geographic data are released. In doing so, SDOs retain autonomy to define the spatial and thematic contexts in which location information is shared and can alter sharing parameters in response to changing needs or trust relationships. Finally, the application permits a digital fingerprinting representation of the sovereign dataset and makes it publicly available as a verifiable record.

Our application furnishes data guardians with a complete approach for accomplishing geospatial data sovereignty on the geoweb. First, during geospatial data sharing SDOs can decide user access levels themselves. Each level in the encrypted volume holds a certain spatial representation, access to which is controlled through sharing a certain portion of the full passphrase. This feature is essential to meet data sovereignty needs effectively (Hojati et al., 2021; Martin & Nissenbaum, 2020). This final volume is shared with different intended parties either directly or through storage on the cloud. Second, having a public record corresponding to this encrypted volume notarised on the blockchain furnishes a trustless approach to data sharing. Since the blockchain is immutable, the authenticity of the shared trust encrypted volume, and the datasets within, can be easily verified using this record during claims and formal litigation. Our complete approach advances existing works and complements similar recent work by McKenzie et al. (2022) who also employ these three security functions in their location-based privacy tool. However, MapSafe solves a very different set of needs and use cases. While *PrivyTo* purely takes a single user location point, our application is aimed at much larger datasets. Moreover, hexagonal binning equips an alternative anonymisation option whereas sharing reduced geospatial data representations provisions a key ingredient for data sovereignty solutions.

The proposed geoprivacy approach can exist alongside the cloud and big data technologies, and protect valuable and sensitive geospatial data against state and international surveillance. The application would empower SDOs to exercise their inherent right to control and govern the circulation, access, and use of geospatial data, contributing to their self-determination, data governance and, ultimately, sovereignty (Reid & Sieber, 2020). Users with a limited GIS skillset can easily and completely safeguard their geospatial data locally in the browser. This level of usability, security and transparency and trust model will increase confidence and encourage SDOs to utilize the application. Useful geospatial data which used to be kept isolated, can now be

shared resulting in vital decision-making. Such applications can help organizations to comply with data privacy laws and regulations, and protect them from legal liability and fines. These are vital for communities that may not have the technical and financial resources to design and support such tools. This approach makes it possible to increase digital sustainability and data sovereignty of the digital space and thus improve the prospects for future generations.

Indeed, combining datasets in anonymization platforms can re-identify subjects and compromise their privacy. Thus, the anonymizer should make appropriate choices with regard to parameters, and verify that the resulting masked datasets are appropriately protected. The browser-based security functions induce a computational delay compared to if they were designed to execute as a plugin, toolbox or a standalone desktop application. Masking time using using Spruill's measure is dependent on the number of points, where 10,000 points are double-masked in 30 s. We accept the limitation that applying this measure for datasets with much more points may not be feasible. However, masking without this measure is much faster—an average of 10,000 points per second. Furthermore, the three-level encryption completes in, on average, 6 s per MB. We do not anticipate sovereign sensitive datasets to contain that many points or be much larger, thus, masking and encryption times are both practical. The subsequent time for notarisation depends on the blockchain transaction queue. This can be sped up by choosing a higher priority. Nevertheless, a link to the URL where the hash will be minted is presented instantly to the user as an assurance.

Although this tool represents a critical first step in geospatial data security, more work needs to be done to cater for all kinds of geospatial data. For instance, masking lines and polygons is still an unresolved challenge in our approach. Another goal is to overcome the browser limitations in the masking and encryption of large geospatial datasets (i.e., beyond 100 Mb). We will explore making the tool available as a desktop application. While these features are planned, in principle, our browser-based tool can still be applied to many sensitive geospatial datasets, since they are point-based, do not have a huge number of points, and are less than 100 MB in size. Moreover, MapSafe can still protect non point-based datasets via encryption and notarisation. The seamless functioning of cloud geospatial data sovereignty deployments will depend on how well existing and future geospatial web systems integrates or interacts with and are supported by data sovereignty tools. MapSafe's JavaScript libraries can easily be integrated as part of existing geospatial web applications, simplifying the design and delivery of geospatial data privacy tools in the future. The sum product of each of these security functions is an application that we believe will increase the adoption of data sovereignty, and therefore improve geospatial data protection within academia and practice.

We anticipate our geospatial data sovereignty tool may also have the potential to aid any process that generates a large volume of point data, in particular, Public Participatory GIS (PPGIS) and Volunteered Geographic Information (VGI) (Goodchild, 2007). New geospatial tools and approaches are central to PPGIS and VGI research, especially web-based tools to using geospatial technologies that empower otherwise marginalized actors and institutions in the spatial decision-making process (Briggs et al., 2020). A browser-based data sovereignty tool will give control to users to safeguard sensitive locations and thus contribute to decision-making. Finally, our approach has the potential to greatly address indigenous data sovereignty (Kukutai & Taylor, 2016) because it represents an enabling system for these communities fine control what and with whom they are willing to share culturally sensitive locations without reliance on a third party (Daly et al., 2019).

6 | CONCLUSION

The growth in the capture of spatial data and the accompanying data sovereignty concerns pose a novel geoprivacy challenge in today's world. Data sovereignty concerns are ever-increasingly intertwined with our digital presence, through increased reliance on cloud-based services and voluntary data storage, which exacerbate privacy and territorial ownership claims. The challenge is how to enable SDOs to utilize these third-party cloud services while simultaneously safeguarding geospatial data when it is shared.

This article presents an application that seeks to reshape or contest technological insecurities at their source—the client computer. MapSafe offers complete geospatial data sovereignty by offering obfuscation, encryption, and notarisation security functions, all executed from the web browser on the client's end, with no sensitive data leaving the computer, except an identifier corresponding to the encrypted data notarised on the blockchain for verification. Furthermore, considerable attention has been paid to accommodating sharing options for this geospatial data, with different obfuscated representations for different trust levels. The proposed solution has been demonstrated to provide adequate usability and security. This approach promises to empower SDOs with control, enhanced trust, transparency, and sovereignty in distributed environments to safeguard their valuable geospatial data without involving any third party. It can be an indispensable tool for resolving place-based conflicts, aiding SDOs to prove their ownership over lands (w/o traditional values), property, and territories.

While these capabilities to protect geographic datasets come at the cost of a set of additional steps, the problems created by not protecting the sovereignty of geospatial data are more detrimental to data protection. Increased reliance on third-party and cloud-based services erodes the user's ability to protect their sovereign data. The utilization of complete data sovereignty tools such as the one presented here offers the potential to take back control and share data with confidence that the appropriate trust relationship is used. Implementation of new data sovereignty technologies may allow SDOs, in general, to increase their resilience at a critical point in human history.

ACKNOWLEDGMENTS

We are grateful for the funding received for this work from Ministry of Business, Innovation and Employment (Ngā Rākau Taketake – Myrtle Rust and Kauri Dieback Research, C09X1817). We are also grateful to Dean Henderson, Cecilia Arienti, and Waitangi Wood for their guidance in this project. We would also like to acknowledge the developers of the open-source libraries that were needed in producing this application. Open access publishing facilitated by The University of Auckland, as part of the Wiley-The University of Auckland agreement via the Council of Australian University Librarians.

CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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ENDNOTES

[1] Shapefiles were sourced from https://mapcruzin.com/free-world-country-arcgis-maps-shapefiles.htm.

[2] A detailed explanation of the nested encryption scheme is provided on the website https://www.mapsafe.xyz/EncDecFlow.pdf.

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How to cite this article: Sharma, P., Martin, M., & Swanlund, D. (2023). MapSafe: A complete tool for achieving geospatial data sovereignty. *Transactions in GIS*, *27*, 1680–1698. https://doi.org/10.1111/tgis.13094

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